So if you think it's time to decide which laptop will work the best for you, depending on your needs and personality, then you'll find the products lined up here seriously fast and stylish. They'll match your lifestyle provided you match theirs—anything between Rs 70,000 to Rs 1,50,000.

Core configuration

There is a minimum level of computing power that is required for most modern applications. If you plan to use the laptop extensively for office productivity-related applications, then you should at least look at a machine that offers you 500 MHz or more computing power. This along with a decent amount of RAM (128 MB) will get you through most applications without a hiccup. The hiccupping starts when you have a poorly configured system (read chipset and processor combination) internally which you will simply have to live with, considering the fact that with laptops you don't have the option of changing motherboard chipsets.

Models from Compaq and ACI were decently configured in terms of pure processing power. Both the Compaq Presario 1722 and the Compaq Presario 1725 came with 1 GHz+ Pentium III processors. In fact, the Compaq Presario 1725 came with the top-of-the-line Tualatin Pentium III-M 1.13 GHz processor. ACI, on the other hand, has all its models configured with at least 1 GHz and above class processors. The ACI Profile 2860 comes fitted with a Celeron 1 GHz and the Profile 3270 comes with a Pentium III-M 1.26 GHz (Tualatin) processor. Also, models from ACI featured GHz class processors, the ACI Profile 3270, and the ACI Xtreme came configured with 1.26 and 1.2 GHz respectively. The ACI Profile 2860 came configured with a Celeron 1 GHz processor.

All the notebooks that we received had decent computing power and did not behave sluggishly, even though some of them had Windows XP Home edition as their standard operating system.

Let's move on to the next critical component in a laptop: RAM. SODIMM (small outline dual inline memory module), as it is called with 144 pin and 3.3 V (instead of regular 168 pin SDRAM and 184 pin DDR-SDRAM), voltage rating is what is used in laptops. RAM is usually placed under the keyboard so that it can be easily removed when upgrading. Sometimes the RAM is placed near the bottom, like the Sony Viao, for easy access. To save space and reduce width of the laptop, the RAM is fixed flat on the board and not vertically as in a desktop.

All the laptops except for the Compaq Armada came configured with at least 128 MB RAM, and some even came configured with 256 MB RAM. Of course, this fact showed in the scores. Two out of the three models from ACI came with 256 MB RAM, and so did the Compaq Presario 1725. The rest were quite stuck in the ordinary computing territory and came configured with 128 MB of RAM.

Mobile CPUs

The CPU is without doubt the most critical component. Its integration into the laptop has major implications as it draws a lot of power, drains resources, and outputs considerable heat. Since space is restricted, adequate cooling while maintaining low noise levels is important. The 'mobile' processor is much smaller (Micro FCPGA2 and FCBGA2 Packaging. These special miniature packages enable thinnest and the lightest notebooks) and has some very nifty instructions built in for conserving power and reducing heat output. For example, the Pentium III uses Enhanced Intel SpeedStep Technology wherein the processor automatically switches between two core frequencies based on CPU demand, optimising battery life and application performance. Another great feature is 'QuickStart' which simply extends battery life by reducing power during pauses in user activity, such as between keystrokes and minor idle states. Deeper Sleep alert state is a dynamic power management mode which is enabled during less than a millisecond of inactivity by the user, such as between keystrokes, thus delivering longer battery life.

The last of the critical component in a laptop is the hard drive. The hard drives of laptops are smaller than that of regular desktops—they are of 2.5-inch rather

VIDEO CARDS FOR LAPTOPS

ATi Radeon Mobility Comes out Trumps!

Just the thought of seeing primitive rendering and lacklustre performance puts off most users when it comes to 3D graphics on a laptop.

The primary reason for dated graphics on laptops was the fact that most laptop manufacturers used chipset solutions that provided no AGP functionality. If included, it would be integrated in the chipset so as to consume less space and generate little heat so that there would be no need for extra cooling. So if you planned to show that amazing rendering in 3D Studio Max on a laptop, you simply couldn't.

Integrated chipset-based shared memory architecture-dependent graphics cards are still the norm for laptops. But the trend is slowly and steadily moving towards better and faster solutions with top 3D graphics chip manufacturing companies showing an increasing interest in this area.

ATi, the world's largest provider of OEM graphics chips for laptops, has released the ATi Radeon Mobility, a chip which came integrated in the ACi Xtreme laptop that we received for testing. The results were simply astounding. Not only was the quality of 2D top notch but the Xtreme had an uncanny knack of running *Quake III* at a whopping speed of 105 fps in normal mode.

The Mobility packs in a single pipeline architecture with the ability to render three texture units in one pass. The ACi Xtreme featured 32 MB of dedicated graphics DDR memory to boot.

So what's the hit, you ask? Well little, if any. The Radeon Mobility has some really cool features to ensure that battery power consumption still remains at low levels by using what ATi calls the 'Voltage/Frequency Modulation' feature. This feature basically allows the chip to run much slower and hence consume less power when the laptop is being used for mundane desktop tasks. But the minute you crank up a 3D application, the Radeon Mobility kicks in, unleashing all of the power that even some desktop PCs don't have.

Features like alpha sub-picture blending and adaptive de-interlacing provide picture-enhancing quality, and other features like iDCT and motion compensation augment performance. You can now finally run cool graphics and games, and render that high polygon count object in 3D Studio Max right on your laptop!